



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Linear Graphs

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- * Drawing Straight Line Graphs
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Your notes

Equations of Straight Lines ($y = mx + c$)

Finding Equations of Straight Lines

What is the equation of a straight line?

- The general **equation** of a **straight line** is $y = mx + c$ where
 - m is the **gradient**
 - c is the **y-intercept**
 - The value where it **cuts** the **y-axis**
- $y = 5x + 2$ is a straight line with
 - gradient 5
 - y-intercept 2
- $y = 3 - 4x$ is a straight line with
 - gradient -4
 - y-intercept 3

How do I find the equation of a straight line from a graph?

- Find the **gradient** by drawing a triangle and using
 - $\text{gradient} = \frac{\text{rise}}{\text{run}}$
 - **Positive** for uphill lines, **negative** for downhill
- Read off the **y-intercept** from the graph
 - Where it cuts the y-axis
- **Substitute** these values into $y = mx + c$

What if no y-intercept is shown?

- If you **can't** read off the **y-intercept**
 - find any **point** on the line
 - **substitute** it into the **equation**



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- **solve to find c**
- For example, a line with gradient 6 passes through (2, 15)
 - The y -intercept is unknown
 - Write $y = 6x + c$
 - Substitute in $x = 2$ and $y = 15$
 - $15 = 6 \times 2 + c$
 - $15 = 12 + c$
 - Solve for c
 - $c = 3$
 - The equation is $y = 6x + 3$

What are the equations of horizontal and vertical lines?

- A **horizontal** line has the equation $y = c$
 - c is the y -intercept
- A **vertical** line has the equation $x = k$
 - k is the x -intercept
- For example
 - $y = 4$
 - $x = -2$

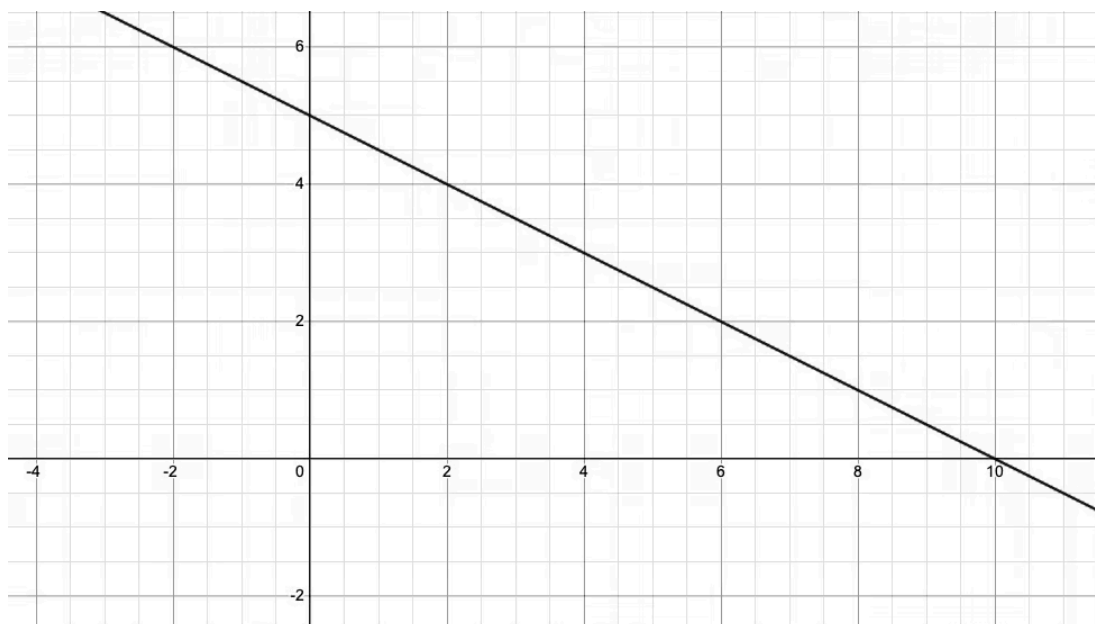


Worked Example

(a) Find the equation of the straight line shown in the diagram below.



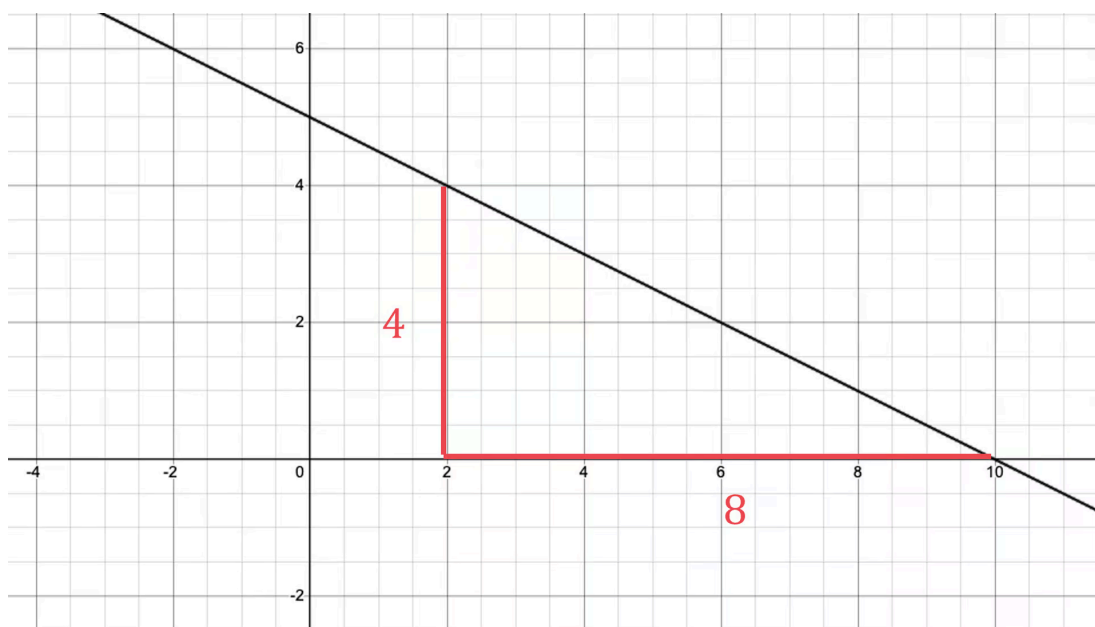
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Find m , the gradient

Identify any two points the line passes through and work out the rise and run

Line passes through $(2, 4)$ and $(10, 0)$





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The rise is 4

The run is 8

Calculate the fraction $\frac{\text{rise}}{\text{run}}$

$$\frac{\text{rise}}{\text{run}} = \frac{4}{8} = \frac{1}{2}$$

The slope is downward (downhill), so it is a negative gradient

$$\text{gradient, } m = -\frac{1}{2}$$

Now find the y-intercept

The line cuts the y-axis at 5

$$\text{y-intercept, } c = 5$$

Substitute the gradient, m , and the y-intercept, c , into $y = mx + c$

$$y = -\frac{1}{2}x + 5$$

(b) Find the equation of the straight line with a gradient of 3 that passes through the point (5, 4).

Substitute $m = 3$ into $y = mx + c$ Leave c as an unknown letter

$$y = 3x + c$$

Substitute $x = 5$ and $y = 4$ into the equationSolve the equation to find c

$$4 = 3 \times 5 + c$$

$$4 = 15 + c$$

$$-11 = c$$

You now know c Replace c with -11 to complete the equation of the line

$$y = 3x - 11$$



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Drawing Straight Line Graphs

Drawing Linear Graphs

How do I draw a straight line from a table of values?

- You may be given a **table of values** with **no** equation
- Use the x and y values to form a point with **coordinates** (x, y)
 - Then **plot** these points
 - Use a ruler to draw a **straight line** through them
 - All points should lie on the line
- For example
 - The points below are $(-3, 0)$, $(-2, 2)$, ... etc

x	-3	-2	-1	0	1	2	3
y	0	2	4	6	8	10	12

How do I draw a straight line using $y = mx + c$?

- Use the **equation** to create your own **table of values**
 - Choose points that are **spread out** across the axes given
- For example, plot $y = 2x + 1$ on axes from $x = 0$ to $x = 10$
 - Substitute in $x = 0$, $x = 5$ and $x = 10$ to get y coordinates
 - Then plot those points

x	0	5	10
y	1	11	21

How do I draw a straight line without using a table of values?

- Assuming the equation is in the form $y = mx + c$



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- **Start** at the **y-intercept**, c
- Then, for every **1 unit** to the **right**, go up **m units**
 - m is the gradient
 - If m is **negative**, go **down**
 - If m is a **fraction**, remember that gradient is change in y divided by change in x
 - A gradient of $\frac{a}{b}$ would be **a** units up for every **b** units right
- This creates a sequence of **points** which you can then join up
 - Be **careful** of counting squares if axes have **different scales**
 - 1 unit might not be 1 square

What if the equation is not in the form $y = mx + c$?

- Equations will not always be presented in the form $y = mx + c$
- **Rearranging** to $y = mx + c$ will make **plotting** these graphs **easier**
- Consider the equation $3x + 5y = 30$
 - Subtract $3x$ from both sides
 - $5y = -3x + 30$
 - Divide both sides by 5
 - $y = -\frac{3}{5}x + 6$
 - It can now be seen that the gradient is $-\frac{3}{5}$ and the y -intercept is 6
- Make sure you only have 1 y on one side, rather than say, $5y$

How can I plot equations in the form $ax + by = c$?

- Instead of rearranging, equations in the form $ax + by = c$, like the example above, can also be plotted by **considering the x and y intercepts** instead
 - Substitute in $x = 0$ to find the y -intercept
 - Substitute in $y = 0$ to find the x -intercept

- E.g. for $3x + 5y = 30$
 - When $x = 0$, $5y = 30$, so $y = 6$
 - When $y = 0$, $3x = 30$, so $x = 10$
- The points (0, 6) and (10, 0) can then be plotted and joined with a straight line



Examiner Tips and Tricks

- Always plot at least 3 points (just in case one of your end points is wrong!)



Worked Example

On the same set of axes, draw the graphs of $\frac{y+1}{3} = x$ and $y = -\frac{3}{5}x + 3$.

Rearrange $\frac{y+1}{3} = x$ into the form $y = mx + c$ to make it easier to plot

$$\begin{aligned}\frac{y+1}{3} &= x \\ y+1 &= 3x \\ y &= 3x-1\end{aligned}$$

For $y = 3x - 1$, create a table of values

x	0	1	2
y	-1	2	5

Plot the points (0, -1), (1, 2) and (2, 5)

Connect with a straight line

Alternatively, start at the y-intercept (0, -1) and mark the next points 3 units up for every 1 unit to the right



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For $y = -\frac{3}{5}x + 3$, create a table of values

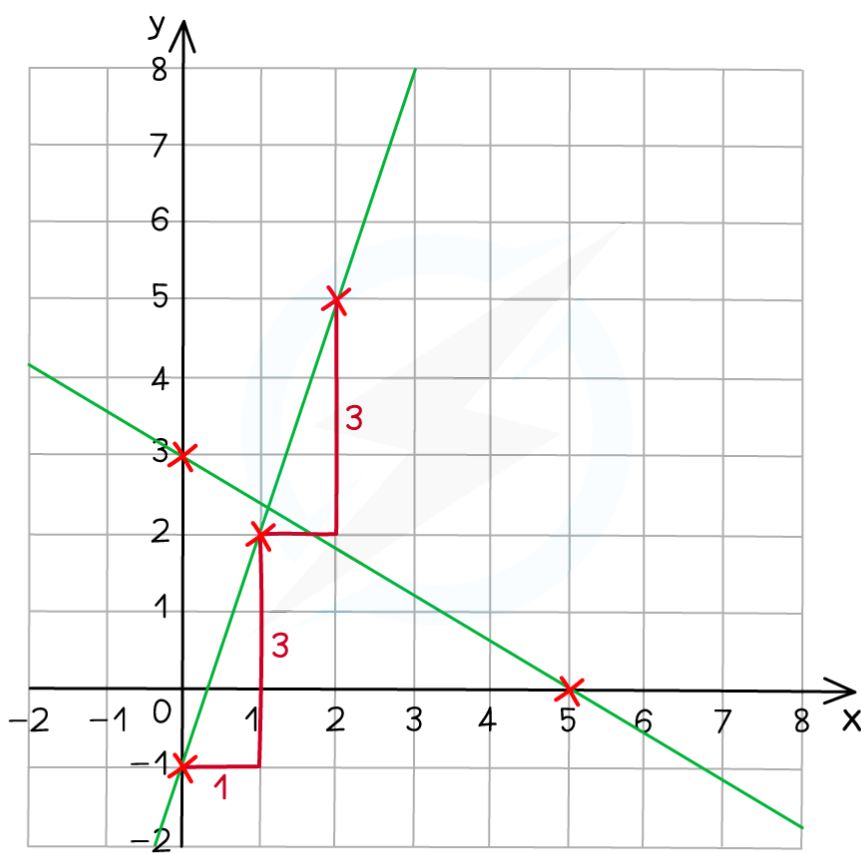
Because of the fraction, $x = 5$ is a good point to include

x	0	3	5
y	3	1.2	0

Plot the points (0, 3), (3, 1.2) and (5, 0)

Connect with a straight line

Alternatively, start at the y-intercept (0, 3) and mark the next points 3 units down for every 5 units to the right



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Your notes

Parallel Lines

Parallel Lines

What are parallel lines?

- Parallel lines are **straight lines** with the **same gradient**
 - Two parallel lines will **never meet**
 - They just stay side-by-side forever
- The **equation** of the line **parallel** to $y = mx + c$ is $y = mx + d$
 - $y = 2x + 1$ and $y = 2x + 5$ are parallel
 - $y = 2x + 1$ and $y = 3x + 1$ are **not** parallel

How do I find the equation of a parallel line?

- For example, to find the **equation** of the line **parallel** to $y = 2x + 1$ which **passes through the point** (3, 14)
 - write the parallel line as $y = 2x + d$
 - using the same gradient of 2
 - substitute** $x = 3$ and $y = 14$ into this equation
 - $14 = 2 \times 3 + d$
 - $14 = 6 + d$
 - solve to find d**
 - $d = 8$
 - The equation is $y = 2x + 8$



Worked Example

Find the equation of the line that passes through the point (2, 1), which is parallel to the straight line $y = 3x + 7$.

Parallel means the gradient will be the same
Use $y = mx + d$ where $m = 3$



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$$y = 3x + d$$

Substitute in $x = 2$ and $y = 1$

$$1 = 3 \times 2 + d$$

Simplify the equation

$$1 = 6 + d$$

Solve the equation to find d (by subtracting 6 from both sides)

$$-5 = d$$

Write out the final answer in the form $y = mx + d$

$$y = 3x - 5$$



Your notes

Perpendicular Lines

Perpendicular Lines

What are perpendicular lines?

- **Perpendicular lines** are **straight lines** which meet at **right-angles** (90°)
- One line may be referred to as a **normal** to the other line

How are the gradients of perpendicular lines related?

- Gradients m_1 and m_2 are **perpendicular** if $m_1 \times m_2 = -1$
 - For example
 - 1 and -1
 - $\frac{1}{3}$ and -3
 - $-\frac{2}{3}$ and $\frac{3}{2}$
- The two gradients are **negative reciprocals** of one another
- We can use $m_2 = -\frac{1}{m_1}$ to find a perpendicular gradient

How can I tell if two lines are perpendicular?

- Given two lines in the form $y = mx + c$, simply check if their gradients (m) are **negative reciprocals** of one another
 - $y = \frac{1}{3}x + 10$ and $y = -3x - 18$ are **perpendicular**
 - $y = \frac{1}{7}x + 16$ and $y = 7x - 8$ are **not perpendicular**
- One or both of the equations may not be written in the form $y = mx + c$
 - In this case, you should **rearrange both equations into the form $y = mx + c$**

- Their **gradients** can then be easily **compared**

How do I find the equation of a line perpendicular to another?

- You need to be able to find the **equation** of line that **passes through a particular point** and is **perpendicular** to another line
 - E.g. $5y = 4x + 30$ which **passes through the point** (8, 3)
- **Rearrange** the equation into the form $y = mx + c$ so that its gradient can be identified more easily
 - $y = \frac{4}{5}x + 6$
- Find the **gradient of the perpendicular** line
 - The gradient of the original line is $\frac{4}{5}$
 - Therefore the **gradient of the perpendicular** line is $-\frac{5}{4}$
 - The perpendicular line has an equation in the form $y = -\frac{5}{4}x + c$
- **Substitute the given point** into the equation for the perpendicular and **solve for c**
 - Substitute (8, 3), into $y = -\frac{5}{4}x + c$
 - $3 = -\frac{5}{4}(8) + c$
 - $c = 13$
- Substitute the value of **c** to find the **equation of the perpendicular**
 - The **equation** of the perpendicular line is $y = -\frac{5}{4}x + 13$
 - This could also be written as $4y = -5x + 52$ or equivalent



Your notes





Your notes

Worked Example

The line L has equation $y - 2x + 2 = 0$.

Find the equation of the line perpendicular to L which passes through the point $(2, -3)$.

Leave your answer in the form $ax + by + c = 0$ where a , b and c are integers.

Rearrange L into the form $y = mx + c$ so we can identify the gradient

$$y - 2x + 2 = 0$$

$$y = 2x - 2$$

Gradient of L is 2

The gradient of the line perpendicular to L will be the negative reciprocal of 2

$$m = -\frac{1}{2}$$

Substitute the point $(2, -3)$ into the equation $y = -\frac{1}{2}x + c$

Solve for c

$$y = -\frac{1}{2}x + c$$

$$-3 = -\frac{1}{2}(2) + c$$

$$-3 = -1 + c$$

$$c = -2$$

Write the full equation of the line

$$y = -\frac{1}{2}x - 2$$

The question asks for the line to be written in the form $ax + by + c = 0$ where a , b and c are integers

Move all the terms to the left hand side

$$\frac{1}{2}x + y + 2 = 0$$

Then multiply every term by 2, to ensure they are all integers

$$x + 2y + 4 = 0$$



Your notes

How do I find the equation of a perpendicular bisector?

- A perpendicular bisector of a line segment cuts the line segment **in half** at a **right angle**
- Finding the **equation of the perpendicular bisector** of a line segment is very similar to finding the equation of a any perpendicular
 - Find the coordinates of the **midpoint** of the line segment
 - The perpendicular bisector will pass through this point
 - Find the **gradient of the line segment**
 - Then find the **negative reciprocal** of this gradient
 - This will be the **gradient of the perpendicular bisector**, m
 - Write the **equation** of the perpendicular bisector in the form $y = mx + c$
 - **Substitute** the **midpoint** of the line segment into the **equation of the perpendicular bisector**
 - **Solve** to find c
 - Write the full **equation** of the perpendicular bisector in the form $y = mx + c$
 - **Rearrange** the equation if the question requires a different form



Worked Example

Find the equation of the perpendicular bisector of the line segment joining the points (4, -6) and (8, 6).

Find the coordinates of the **midpoint** of the line segment
The perpendicular bisector will pass through this point

$$\left(\frac{4+8}{2}, \frac{-6+6}{2} \right) = (6, 0)$$

Find the gradient of the line segment



Your notes

$$\frac{-6-6}{4-8} = \frac{-12}{-4} = 3$$

Find the negative reciprocal of this

This will be the gradient of the perpendicular bisector, m

$$m = -\frac{1}{3}$$

Write the equation of the perpendicular bisector in the form $y = mx + c$

$$y = -\frac{1}{3}x + c$$

Substitute in the midpoint (6, 0) and solve to find c

$$0 = -\frac{1}{3}(6) + c$$

$$0 = -2 + c$$

$$c = 2$$

Write the full equation of the perpendicular bisector

$$y = -\frac{1}{3}x + 2$$